

A Learning-Based Framework for Inverse Problems using Dirichlet and Neumann Boundary Data

Dabin Park¹ Sanghyun Lee² Sunghwan Moon¹

¹Kyungpook National University

²Florida State University

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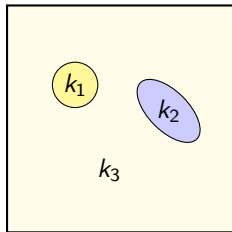
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Elliptic PDE

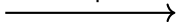
$$\begin{cases} -\nabla \cdot (k(\mathbf{x}) \nabla p(\mathbf{x})) &= f(\mathbf{x}) & \mathbf{x} \in \Omega, \\ p(\mathbf{x}) &= p_D(\mathbf{x}) & \mathbf{x} \in \partial\Omega, \end{cases}$$

- Ω is an open bounded domain in $\mathbb{R}^2$.
- f is the source function.
- p_D is the Dirichlet boundary condition.
- $k > 0$.

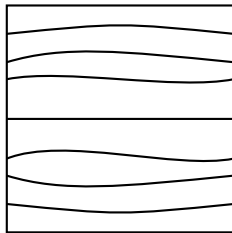
Coefficient (k_i) is known



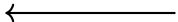
Forward problem



Solve the PDE



Inverse problem

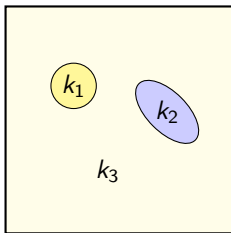


Solution p is known

Forward problem: Given p_D , k , f , solve the PDE to obtain the solution p .

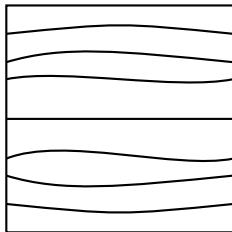
Inverse problem: Given p_D , p , f , reconstruct the coefficient k .

Reconstruct the coefficient (k_i)



Inverse problem
←

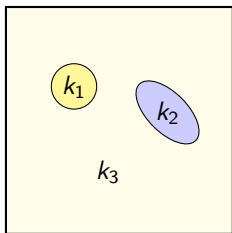
Solution p is known



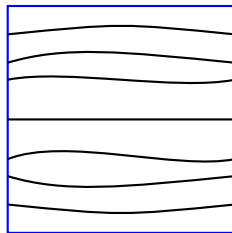
Inverse problem: Given p_D , p , f , reconstruct the coefficient k .

Reconstruct the coefficient (k_i)

Boundary measurement is known



Inverse problem
←

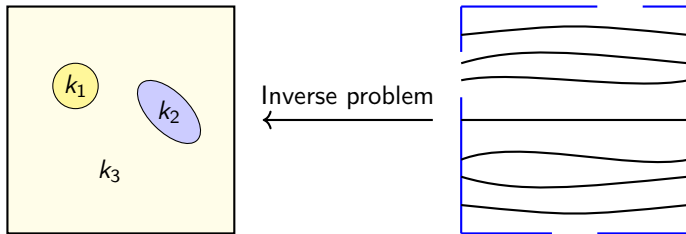


Inverse problem: Given p_D , $\emptyset$, f , reconstruct the coefficient k .

Our inverse problem: Given p_D , $k \partial_\nu p|_{\partial\Omega}$, f , reconstruct the coefficient k .

Reconstruct the coefficient (k_i)

Partial boundary measurement is known

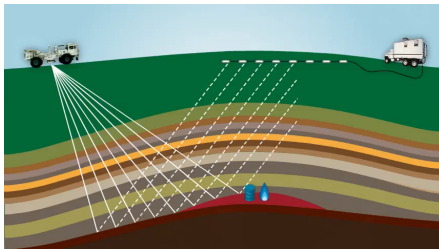
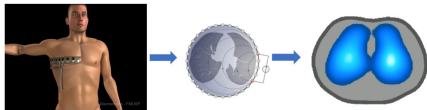


Inverse problem: Given p_D , ϕ , f , reconstruct the coefficient k .

Our inverse problem: Given $p_D|_{\Gamma}$, $k \partial_{\nu} p|_{\Gamma}$, $f|_{\Gamma}$, reconstruct the coefficient k , where $\Gamma \subset \partial\Omega$.

Application

Simple. Non Invasive. Bedside.



Dirichlet-to-Neumann (DtN) map

$$\Lambda_k : H^{1/2}(\partial\Omega) \rightarrow H^{-1/2}(\partial\Omega),$$
$$p_D \mapsto h = k\partial_\nu p|_{\partial\Omega}$$

- p is the solution of an Elliptic PDE with the given Dirichlet boundary condition p_D .
- ν is the unit outward normal vector on $\partial\Omega$.
- $H^r(\partial\Omega)$ is the Sobolev space of order r , where $r \in \mathbb{R}$.

Let $\Gamma \subset \partial\Omega$ be a non-empty open subset of the boundary.
The following Theorem shows the uniqueness of the coefficient k .

Theorem (Corollary 1.19 in G. Uhlmann (2014))

Let $\Omega \subset \mathbb{R}^2$ be a bounded domain with a smooth boundary, and let $k_1, k_2 \in C^{4+\alpha}(\overline{\Omega})$ be non-vanishing functions for some $\alpha > 0$. Assume that

$$\Lambda_{k_1}(p_D) = \Lambda_{k_2}(p_D) \text{ on } \Gamma \text{ for all } p_D \in H^{1/2}(\partial\Omega), \text{ supp } p_D \subset \Gamma.$$

Then, $k_1 = k_2$ in Ω .

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Remark

- If $\Lambda_{k_1} = \Lambda_{k_2}$ on Γ , then $k_1 = k_2$ in Ω .

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Remark

- If $\Lambda_{k_1} = \Lambda_{k_2}$ on Γ , then $k_1 = k_2$ in Ω .
- This result justifies our approach: reconstructing the coefficient k from boundary measurements is theoretically valid.

Let ω be either Γ or $\partial\Omega$.

Cauchy data—pairs of Dirichlet and Neumann boundary measurements:

$$\left\{ (p_D, h) : p_D \in H^{1/2}(\partial\Omega), \text{ supp } p_D \subset \omega, h = k\partial_\nu p|_\omega \right\}.$$

These data pairs serve as practical replacement for the DtN map.

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These data pairs serve as practical replacement for the DtN map.

Problem

Given a sufficiently large finite collection of Cauchy data:

$$\left\{ (p_{D,i}, h_i)_{i=1,\dots,N} : p_{D,i} \in H^{1/2}(\partial\Omega), \text{ supp } p_{D,i} \subset \omega, h_i = k\partial_\nu p_i|_\omega \right\},$$

the coefficient k can be numerically reconstructed in Ω .

Learning-Based Framework

Deep Learning

Finite Element Method

..... : Data flow

→ : Computation results

Input

- Domain points

$$S = \{\mathbf{x} \mid \mathbf{x} \in \overline{\Omega}\}$$

Learning-Based Framework

Deep Learning

Finite Element Method

..... : Data flow
→ : Computation results

Input

- Domain points
 $S = \{\mathbf{x} \mid \mathbf{x} \in \bar{\Omega}\}$

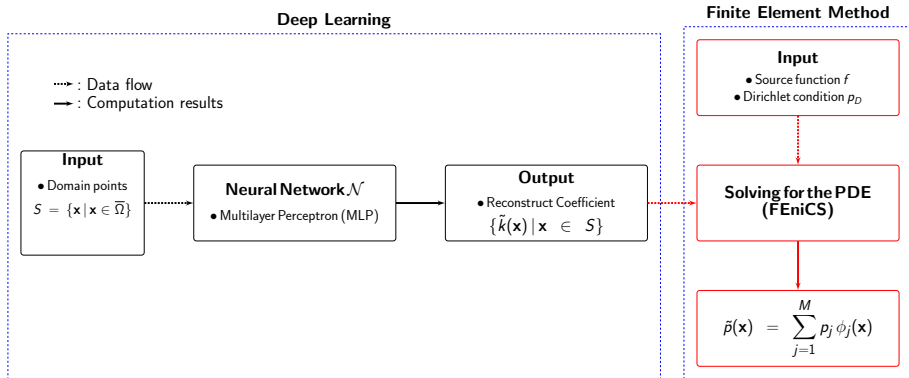
Neural Network $\mathcal{N}$

- Multilayer Perceptron (MLP)

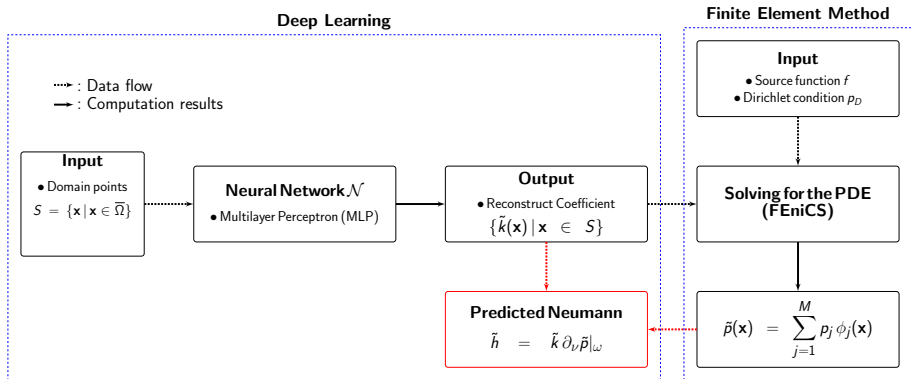
Output

- Reconstruct Coefficient
 $\{\tilde{k}(\mathbf{x}) \mid \mathbf{x} \in S\}$

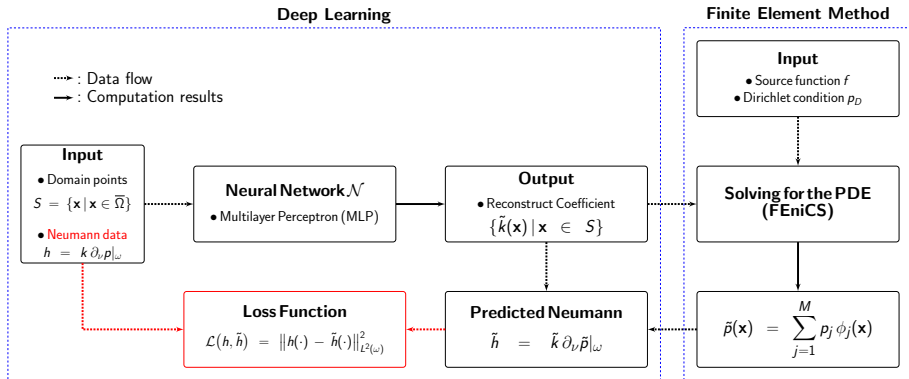
Learning-Based Framework



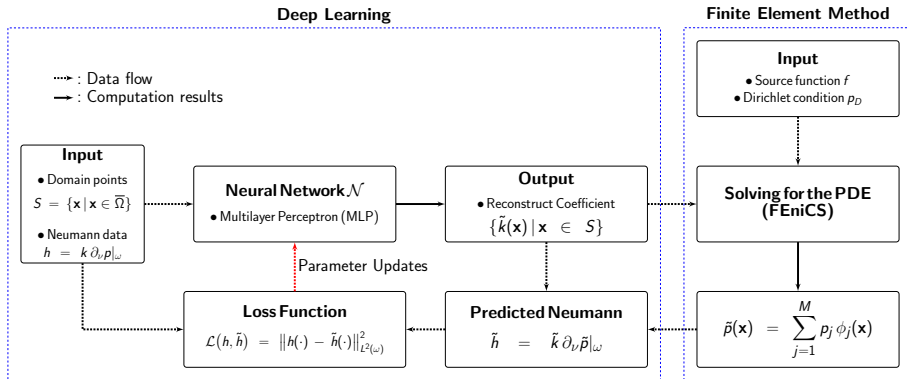
Learning-Based Framework



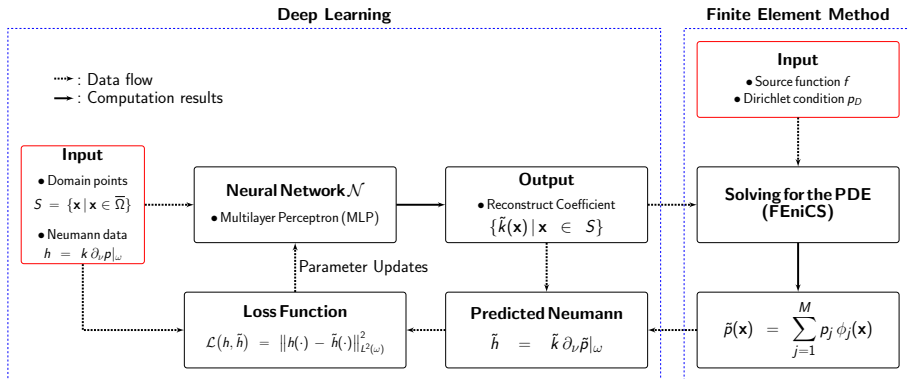
Learning-Based Framework



Learning-Based Framework



Learning-Based Framework



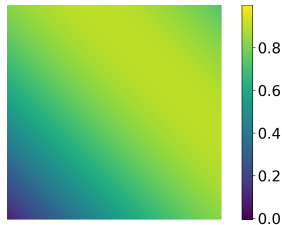
- Only Cauchy data (Dirichlet + Neumann) used
- No exact k needed
- Useful in real-world scenarios

The loss function

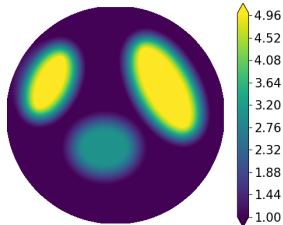
$$\mathcal{L}(\mathcal{D}, \tilde{\mathcal{D}}) = \sum_{i=1}^N \|h_i - \tilde{k} \partial_\nu \tilde{p}_i\|_{L^2(\omega)}^2,$$

- $\mathcal{D} = \{(p_{D,i}, h_i)\}_{i=1}^N$ is the training Cauchy data.
- $\tilde{\mathcal{D}} = \{(p_{D,i}, \tilde{k} \partial_\nu \tilde{p}_i|_\omega)\}_{i=1}^N$ is the predicted Cauchy data.
- $\|\cdot\|_{L^2(\omega)}$ denotes the L^2 norm on ω .

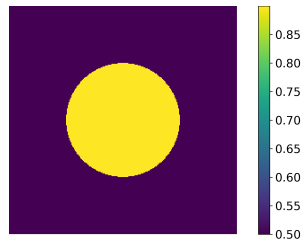
Numerical Results



(a) Continuous

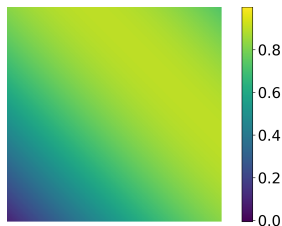


(b) Phantom

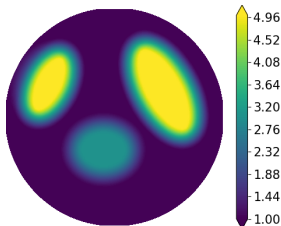


(c) Discontinuous

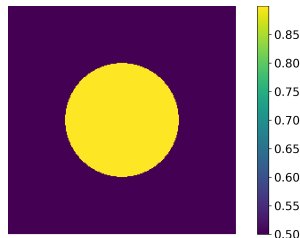
Numerical Results



(a) Continuous



(b) Phantom

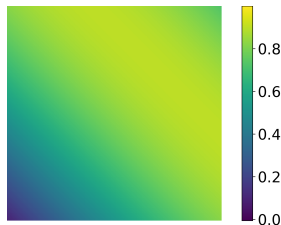


(c) Discontinuous

Experiments Setup

- The Dirichlet boundary condition is $p_D = \cos(a\pi x) \cos(b\pi y)|_{\omega}$. Here, a and b are independently sampled from a uniform distribution over $[0, 2]$.

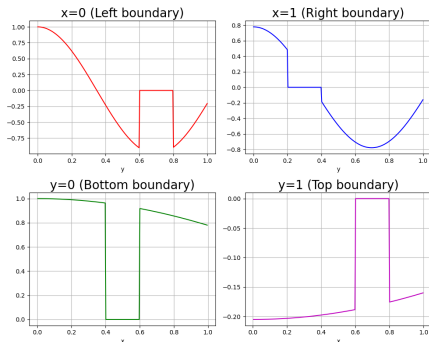
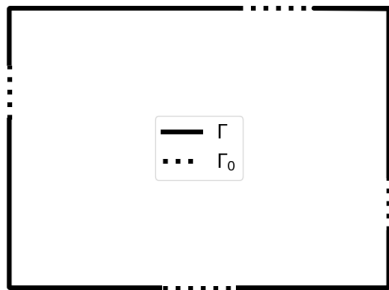
Example 1: Continuous Coefficient

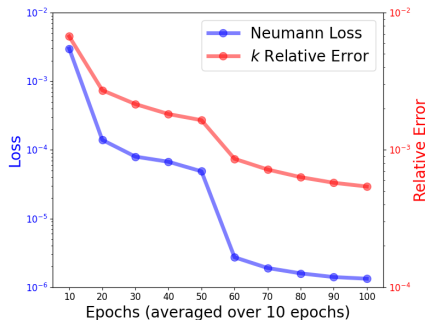


- The unit square domain.
- Fully observed: $\omega = \partial\Omega$.
- Partially observed: $\omega = \Gamma$.

In the partially observed setting:

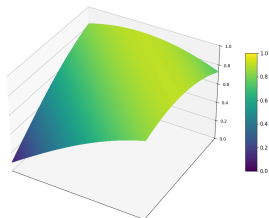
- Γ is the observation region, which covers 80% of the boundary $\partial\Omega$.
- $\Gamma_0 = \partial\Omega \setminus \Gamma$.
- On Γ , the Dirichlet boundary condition and the source function are the same as in the fully observed setting.
- On Γ_0 , $p_D = 0$ and $f = 0$.



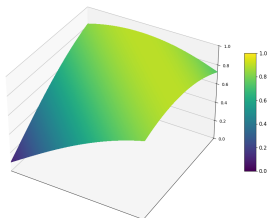


The relative error:

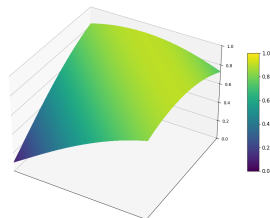
$$\|\tilde{k}\|_{\text{rel}} = \frac{\|k - \tilde{k}\|_{L^2(\overline{\Omega})}}{\|k\|_{L^2(\overline{\Omega})}}.$$



(a) exact k

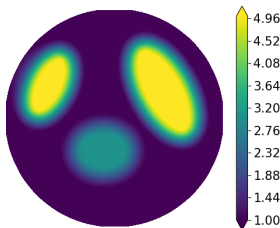


(b) reconstructed $\tilde{k}$
in fully observed



(c) reconstructed $\tilde{k}$
in partially observed

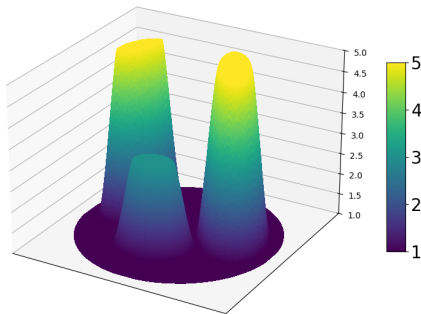
Example 2: Phantom Coefficient



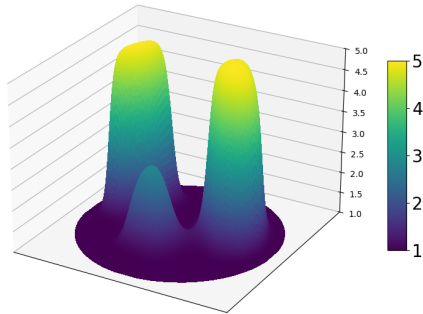
- Fully observed, circle domain.
- The loss function includes a regularization term:

$$\sum_{i=1}^N \|h_{g_i} - \tilde{k} \partial_\nu \tilde{p}_{g_i}\|_{L^2(\partial\Omega)}^2 + \lambda \|\nabla \tilde{k}\|_{L^2(\Omega)}^2.$$

- Post-processing: the hyperbolic tangent function is applied to constrain $\tilde{k}$ to the range $[2.0, 5.0]$.

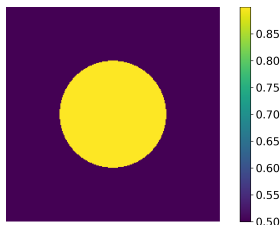


(a) exact k



(b) reconstructed $\tilde{k}$

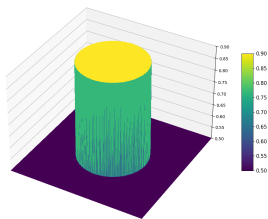
Example 3: Discontinuous Coefficient



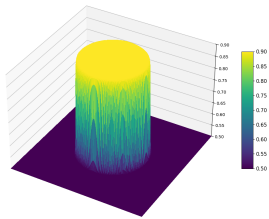
- Fully observed, unit square domain.
- The loss function includes a regularization term:

$$\sum_{i=1}^N \|h_{g_i} - \tilde{k} \partial_\nu \tilde{p}_{g_i}\|_{L^2(\partial\Omega)}^2 + \lambda \|\nabla \tilde{k}\|_{L^2(\Omega)}^2.$$

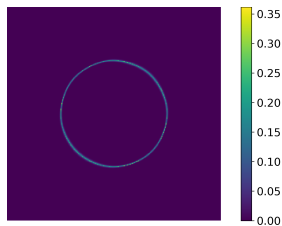
- Post-processing: the hyperbolic tangent function is applied to constrain $\tilde{k}$ to the range $[0.5, 0.9]$.



(a) exact k



(b) reconstructed $\tilde{k}$



(c) absolute error $|k - \tilde{k}|$

Conclusion

1. We proposed a learning-based framework to reconstruct the unknown coefficient k in elliptic PDE.
2. The framework uses only Cauchy data (Dirichlet and Neumann pairs) to reconstruct the coefficient.
3. Numerical results confirm the effectiveness of our framework in both smooth and discontinuous cases.

References



G. Uhlmann, Inverse problems: seeing the unseen, *Bulletin of Mathematical Sciences* 4 (2014) 209–279.

Thank you!

Email: ekqls5264@knu.ac.kr



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